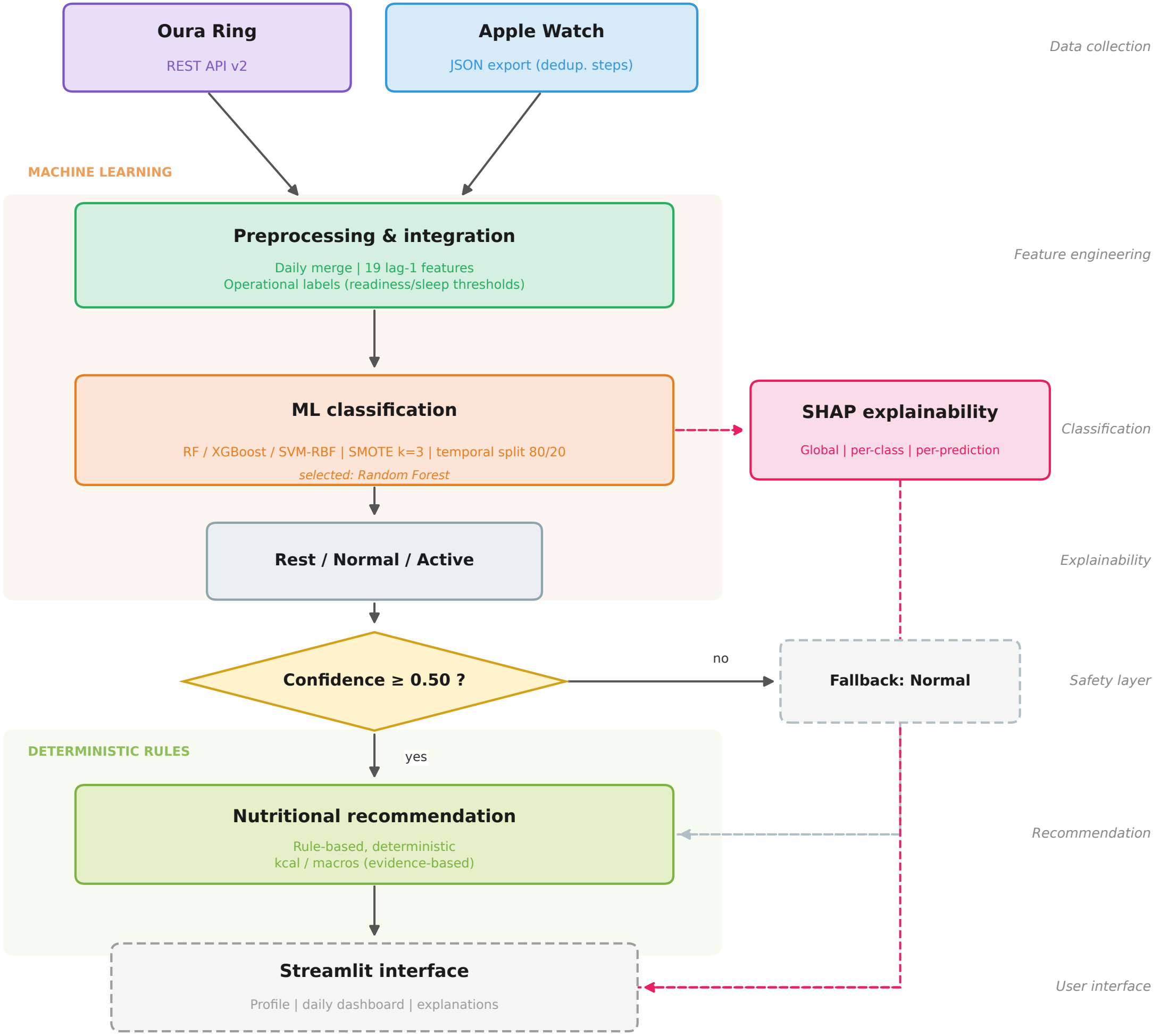

Oura Ring
REST API v2
Apple Watch
JSON export (dedup. steps)
Data collection
MACHINE LEARNING
Preprocessing & integration
Daily merge | 19 lag-1 features
Operational labels (readiness/sleep thresholds)
Feature engineering
ML classification
RF / XGBoost / SVM-RBF | SMOTE k=3 | temporal split 80/20
selected: Random Forest
SHAP explainability
Global | per-class | per-prediction
Classification
Rest / Normal / Active
Explainability
Confidence ≥ 0.50 ?
no
Fallback: Normal
Safety layer
DETERMINISTIC RULES
yes
Nutritional recommendation
Rule-based, deterministic
kcal / macros (evidence-based)
Recommendation
Streamlit interface
Profile | daily dashboard | explanations
User interface